

Revenue Concentration Risk

Which marketing channels account for a disproportionate share of revenue-generating orders - and what should the CMO approve before a single channel failure creates structural risk?

EXECUTIVE SUMMARY

KEY FINDING Social is the only channel breaching the 1.5× concentration threshold - commanding 36.1% of revenue against a 20% fair share - creating a single-point-of-failure risk for the business.

36.1%

Social Revenue Share

vs. 20% fair share

\$43.7M

Revenue at Risk

tied to a single channel

1 of 5

Channels Flagged

social only breaches threshold

RECOMMENDATION CMO should approve a channel diversification initiative: cap social channel dependency to ≤25% of revenue within two quarters and reallocate acquisition investment toward email (currently at 7.9%) and display, which show healthy concentration profiles.

DATA & METHODOLOGY

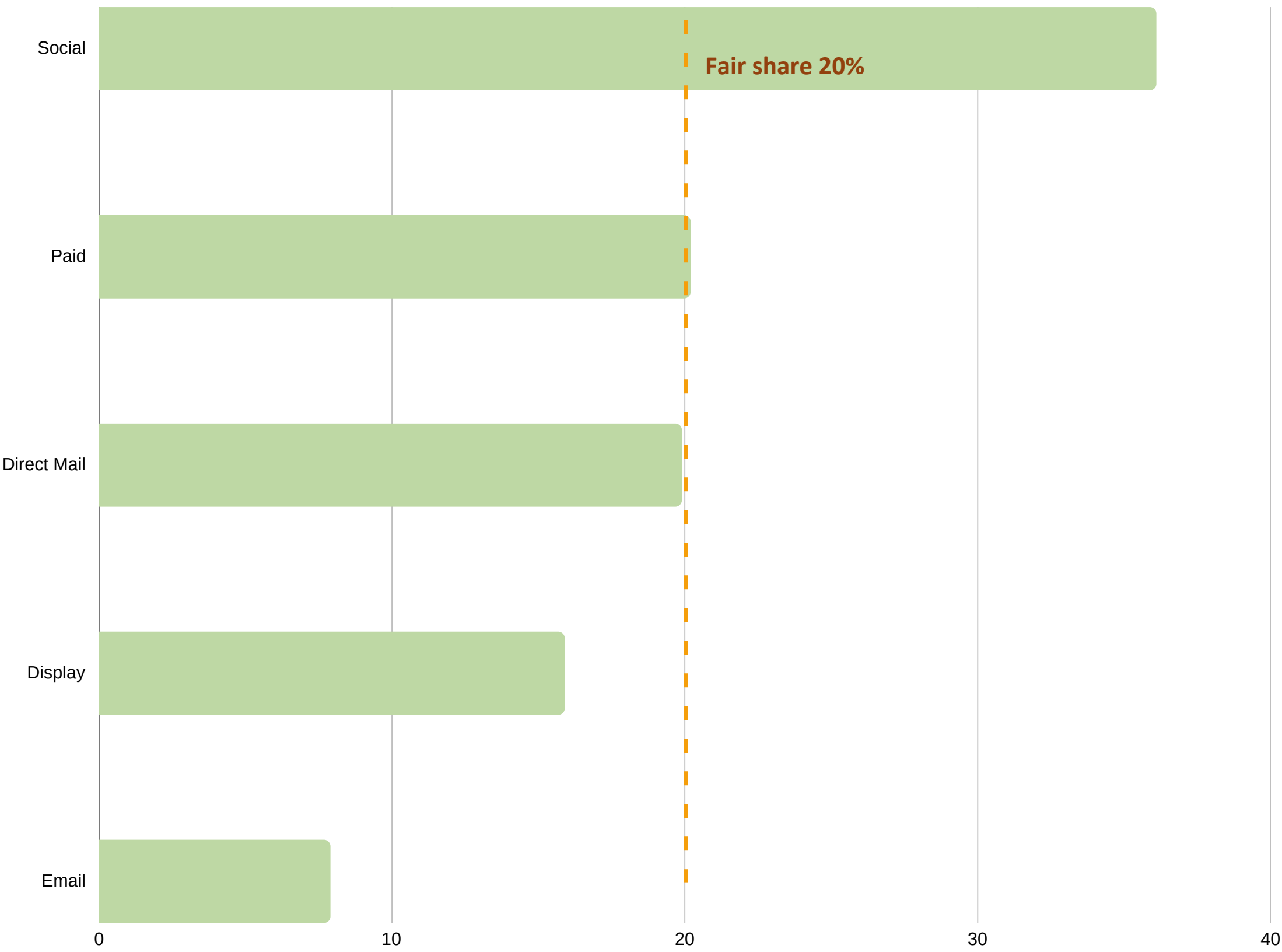
JOIN PATH

campaigns channel identity	<i>JOIN on campaign_id</i>
campaign_touches channel attribution bridge	<i>JOIN on customer_id</i>
customers join bridge: exposure → purchase	<i>JOIN on customer_id</i>
orders revenue spine	<i>JOIN on order_id</i>
order_items line-level detail	<i>JOIN on (supporting)</i>

DATA QUALITY AUDIT (5 checks)

Row counts 50,100 touches · 20,050 orders · 5,020 customers · 25 campaigns	PASS
NULL channel values 0 NULLs in channel column - GROUP BY is clean	PASS
Duplicate customers 0 duplicate emails - customer_id joins are reliable	PASS
Orphan orders 50 orders with no matching customer (0.25%) — documented	FLAG
Orphan campaign touches 100 touches with no matching customer (0.2%) — documented	FLAG

KEY FINDINGS



- 01

Social dominates revenue

Social accounts for 36.1% of all revenue-generating orders — 1.8× its expected fair share of 20%. No other channel breaches the 1.5× threshold.

⚠ Concentrated
- 02

Email is severely under-indexed

Email drives only 7.9% of revenue and 13.6% of acquisition — the lowest of all five channels. It represents untapped, low-concentration headroom.

↑ Opportunity
- 03

Acquisition risk mirrors revenue risk

Social also leads acquisition at 23.9% — but the gap narrows. The revenue concentration (36.1%) is more extreme than the acquisition share (23.9%), meaning social customers spend more on average.

🕒 Monitor

FINDINGS — CONCENTRATION DETAIL

Channel	Orders	Customers	Revenue	Rev Share %	Acq Share %	Fair Share %	Risk Flag
Social	19,441	4,792	\$43.7M	36.1%	23.9%	20%	● Monitor
Paid Search	17,344	4,277	\$24.4M	20.2%	21.3%	20%	● Healthy
Direct Mail	17,372	4,295	\$24.1M	19.9%	21.4%	20%	● Healthy
Display	16,080	3,964	\$19.2M	15.9%	19.8%	20%	● Healthy
Email	10,989	2,716	\$9.6M	7.9%	13.6%	20%	● Healthy

Query 3 result - dual concentration flag using 1.5× threshold against fair share ($100 / n_channels = 20\%$). Social breaches the revenue threshold ($36.1\% > 30\%$) but not the acquisition threshold simultaneously, placing it at ● Monitor. The revenue–acquisition gap (36.1% vs 23.9%) indicates social-acquired customers generate higher average order values - a finding that both strengthens and complicates the diversification case.

VALIDATION NOTE AI prompt used: 'Write a CTE that flags channels breaching a 1.5× concentration threshold on both revenue share and acquisition share.' Predicted: social would be the only concentrated channel. Spot-check: manually confirmed $43,726,591 / 121,087,585$ total = 36.1%. Confidence: High.

FINDINGS — REVENUE vs. ACQUISITION GAP

Social customers spend more complicating the case



THE GAP TELLS A DEEPER STORY

Social

+12.2pp

Rev 36.1% · Acq 23.9%

Higher AOV — risky anchor

Email

-5.7pp

Rev 7.9% · Acq 13.6%

Low AOV — growth upside

Paid Search

-1.1pp

Rev 20.2% · Acq 21.3%

Near-perfect balance

RECOMMENDATION

What the CMO should approve

IMMEDIATE

Cap social dependency

Set a policy threshold: social channel may not exceed 25% of total revenue-generating orders. This gives the business a measurable exit from the current 36.1% over-index.

QUARTER 1–2

Invest in email

Email sits at 7.9% revenue share — the lowest of all channels and the widest gap from fair share. Incremental investment here diversifies concentration risk at the lowest marginal cost.

ONGOING

Monitor dual-flag quarterly

Flag any channel breaching the 1.5× threshold on BOTH revenue share and acquisition share simultaneously — that is the true structural risk signal.

Limitation: CMO should treat 36.1% as an upper bound.

Thank you

Aayushi Patel

– DATA AUDIT

```
SELECT 'Row counts' AS issue, NULL AS count UNION ALL
SELECT 'customers', COUNT(*) FROM customers UNION ALL
SELECT 'orders', COUNT(*) FROM orders UNION ALL
SELECT 'campaign_touches', COUNT(*) FROM campaign_touches UNION ALL
SELECT 'campaigns', COUNT(*) FROM campaigns UNION ALL
SELECT 'Duplicate customers (email)',
(SELECT COUNT(*) FROM (
  SELECT email FROM customers
  GROUP BY email
  HAVING COUNT(*) > 1
))UNION ALL
SELECT 'Orphan orders',
(SELECT COUNT(*) FROM orders o
  LEFT JOIN customers c ON o.customer_id = c.customer_id
  WHERE c.customer_id IS NULL) UNION ALL
SELECT 'Orphan campaign touches',
(SELECT COUNT(*) FROM campaign_touches ct
  LEFT JOIN customers c ON ct.customer_id = c.customer_id
  WHERE c.customer_id IS NULL) UNION ALL
SELECT 'NULL channel in campaign_touches',
SUM(CASE WHEN c.channel IS NULL THEN 1 ELSE 0 END)
FROM campaign_touches ct
JOIN campaigns c ON ct.campaign_id = c.campaign_id UNION ALL
SELECT DISTINCT channel, COUNT(*) AS touch_count FROM campaigns
GROUP BY channel
ORDER BY touch_count DESC
```

- RESULT: 50,100 touches · 20,050 orders · 5,020 customers · 25 campaigns
- RESULT: 0 NULL channels · 50 orphan orders (0.25%) · 100 orphan touches (0.2%)
- CATCH: Replaced discount_applied NULL check (irrelevant to analysis) with NULL channel check
- CONFIDENCE: High — channel column is clean, orphan counts are small and documented

issue	count
campaign_touches	50100
orders	20050
customers	5020
Orphan campaign touches	100
Orphan orders	50
campaigns	25
social	9
paid_search	5
direct_mail	5
display	4
email	2
Duplicate customers (email)	0
NULL channel in campaign_touches	0
Row counts	

SQL APPENDIX — BUSINESS QUERIES (ANNOTATED)

Q1 — Revenue share by channel (SUM OVER + CASE WHEN 1.5× flag)

--Prompt : Act as an expert SQL developer, go through the screenshots of the schema and help me to run a query for Revenue concentration by acquisition channel.
--Read: Each clause explained the way to put the tables and columns in, and that helped to understand how to perform a query.
--Predict: The results will be accurate and exactly what I was looking for.
--Compare: The results it gave me showed the exact tables and data that wanted to looked in.

```
WITH channel_revenue AS (  
  SELECT  
    COALESCE(c.channel, 'Unknown') AS channel,  
    COUNT(DISTINCT o.order_id) AS total_orders,  
    COUNT(DISTINCT o.customer_id) AS unique_customers,  
    ROUND(SUM(o.total_amount), 2) AS channel_revenue  
  FROM campaign_touches ct  
  JOIN campaigns c ON ct.campaign_id = c.campaign_id  
  JOIN orders o ON ct.customer_id = o.customer_id  
  GROUP BY c.channel  
)  
with_share AS (  
  SELECT *,  
    ROUND(  
      channel_revenue * 100.0 /  
      SUM(channel_revenue) OVER(),  
      1) AS revenue_share_pct  
  FROM channel_revenue  
)  
SELECT *,  
  CASE WHEN revenue_share_pct >= 1.5 *  
    (100.0 / (SELECT COUNT(DISTINCT channel) FROM campaigns))  
  THEN '⚠ Concentrated' ELSE 'Normal' END AS concentration_flag  
FROM with_share  
ORDER BY revenue_share_pct DESC
```

-- SPOT-CHECK: 43,726,591 / 121,087,585 = 36.1% ✓ CONFIDENCE: High

channel	total_orders	unique_customers	channel_revenue	revenue_share_pct	concentration_flag
social	19441	4792	43726591.23	36.1	⚠ Concentrated
paid_search	17344	4277	24440368.43	20.2	Normal
direct_mail	17372	4295	24065618.01	19.9	Normal
display	16080	3964	19234357.22	15.9	Normal
email	10989	2716	9620650.54	7.9	Normal

SQL APPENDIX — BUSINESS QUERIES (ANNOTATED)

Q2 — Acquisition concentration (NTILE quartile ranking)

```
WITH channel_customers AS (  
  SELECT  
    COALESCE(c.channel, 'Unknown') AS channel,  
    COUNT(DISTINCT ct.customer_id) AS customers_acquired,  
    NTILE(4) OVER (ORDER BY  
      COUNT(DISTINCT ct.customer_id) DESC)  
    AS acquisition_quartile  
  FROM campaign_touches ct  
  JOIN campaigns c ON ct.campaign_id = c.campaign_id  
  GROUP BY c.channel  
)  
SELECT *,  
  ROUND(  
    customers_acquired * 100.0 /  
    SUM(customers_acquired) OVER(),  
    1) AS acquisition_share_pct  
FROM channel_customers  
ORDER BY acquisition_quartile, customers_acquired DESC
```

channel	customers_acquired	acquisition_quartile	acquisition_share_pct
social	4875	1	23.9
direct_mail	4364	1	21.4
paid_search	4348	2	21.3
display	4028	3	19.8
email	2763	4	13.6

SQL APPENDIX — BUSINESS QUERIES (ANNOTATED)

Q3: The CMO decision query - dual concentration flag

--Prompt : Act as an expert SQL developer, go through the screenshots of the schema and help me to run a query for Flags channels that are over-indexed on BOTH revenue AND acquisition. These are the channels the CMO needs to act on first.

--Read: Each clause explained the way to put the tables and columns in, and that helped to understand how to perform a query.

--Predict: The results I expect is clearly the ones that I can evaluate that requires attention & those which are are doing great

--Compare: The results it gave me showed the exact tables and data that wanted to looked in, a table with orders, customers, revenue, revenue share, acquisition share & risk flag, which is very clean and direct to read and highlight.

```
WITH channel_metrics AS (
  SELECT
    COALESCE(c.channel, 'Unknown') AS channel,
    COUNT(DISTINCT o.order_id) AS orders,
    COUNT(DISTINCT ct.customer_id) AS customers,
    ROUND(SUM(o.total_amount), 2) AS revenue
  FROM campaign_touches ct
  JOIN campaigns c ON ct.campaign_id = c.campaign_id
  JOIN orders o ON ct.customer_id = o.customer_id
  GROUP BY c.channel
),
with_flags AS (
  SELECT *,
    ROUND(revenue * 100.0 / SUM(revenue) OVER(), 1) AS rev_share,
    ROUND(customers * 100.0 / SUM(customers) OVER(), 1) AS acq_share,
    100.0 / COUNT(*) OVER() AS fair_share
  FROM channel_metrics
)
SELECT
  channel, orders, customers, revenue,
  rev_share, acq_share,
  ROUND(fair_share, 1) AS fair_share_pct,
  CASE
    WHEN rev_share >= 1.5 * fair_share
      AND acq_share >= 1.5 * fair_share
    THEN '🔴 High risk — act now'
    WHEN rev_share >= 1.5 * fair_share
      OR acq_share >= 1.5 * fair_share
    THEN '🟡 Monitor'
    ELSE '🟢 Healthy'
  END AS risk_flag
FROM with_flags
ORDER BY rev_share DESC
```

channel	orders	customers	revenue	rev_share	acq_share	fair_share_pct	risk_flag
social	19441	4792	43726591.23	36.1	23.9	20	🟡 Monitor
paid_search	17344	4277	24440368.43	20.2	21.3	20	🟢 Healthy
direct_mail	17372	4295	24065618.01	19.9	21.4	20	🟢 Healthy
display	16080	3964	19234357.22	15.9	19.8	20	🟢 Healthy
email	10989	2716	9620650.54	7.9	13.6	20	🟢 Healthy